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Amazon Inspector SBOM Generator

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Recommended tasks

Learn about

Previous versions of the Amazon Inspector SBOM Generator (https://docs.aws.amazon.com/inspector/latest/user/sbom-generator-versions.html)

Understand how to generate a software bill of materials (https://docs.aws.amazon.com/inspector/latest/user/sbom-generator-ecosystem-collection.html)

Understand how to generate a software bill of materials (https://docs.aws.amazon.com/inspector/latest/user/sbom-generator-dependency-collection.html)

Understand operating system package collection (https://docs.aws.amazon.com/inspector/latest/user/sbom-generator-operating-system-collection.html)

Understand how to scan SBOM for vulnerabilities (https://docs.aws.amazon.com/inspector/v2/APIReference/API_scan_ScanSbom.html)

How to

Export software bill of materials with Amazon Inspector (https://docs.aws.amazon.com/inspector/latest/user/sbom-export.html)

Integrate Amazon Inspector scans into CI/CD pipeline (https://docs.aws.amazon.com/inspector/latest/user/scanning-cicd.html)

Use Amazon Inspector to scan Dockerfiles (https://docs.aws.amazon.com/inspector/latest/user/dockerfile-checks.html)

Version references (sbom-

Configure and use the InspectorScan action in CodePipeline (https://docs.aws.amazon.com/codepipeline/latest/userguide/action-reference-InspectorScan.html)

Install or update the latest version of the AWS CLI (https://docs.aws.amazon.com/cli/latest/userguide/getting-started-install.html)

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A Software Bill of Materials (SBOM) is a formally structured list of components, libraries, and modules (https://docs.aws.amazon.com/whitepapers/latest/practicing-continuous-integration-continuou delivery/software-bill-of-materials-sbom.html) required to build a piece of software. The Amazon Inspector SBOM Generator (Sbomgen) is a tool that produces an SBOM for archives, container images, directories, local systems, and compiled Go and Rust binaries. Sbomgen scans for files th contain information about installed packages. When Sbomgen finds a relevant file, it extracts package names, versions, and other metadata. Sbomgen then transforms package metadata int CycloneDX SBOM. You can use Sbomgen to generate the CycloneDX SBOM as a file or in STDOU send SBOMs to Amazon Inspector for vulnerability detection. You can also use Sbomgen as part the CI/CD integration (https://docs.aws.amazon.com/inspector/latest/user/scanning-cicd.html) , which container images automatically as part of your deployment pipeline.

Supported packages types

Sbomgen collects inventory for the following package types:

- Alpine APK
- Debian/Ubuntu DPKG
- Red Hat RPM
- C#
- Go
- Java
- Node.js
- PHP
- Python
- Ruby
- Rust

Supported container image configuration checks

Sbomgen can scan standalone Dockerfiles and build history from exisiting images for security iss
For more information, see Amazon Inspector Dockerfile checks (https://docs.aws.amazon.com/inspector/latest/user/dockerfile-checks.html) .

Installing Sbomgen

Sbomgen is only available for Linux operating systems.

You must have Docker installed if you want Sbomgen to analyze locally cached images. Docker is required to analyze images exported as `.tar` files or images hosted in remote container registr

Amazon Inspector recommends that you run Sbomgen from a system with at least the following hardware specs:

- 4x core CPU
- 8 GB RAM

To install Sbomgen

1. Download the latest Sbomgen zip file from the correct URL for your architecture:

   Linux AMD64: https://amazon-inspector-sbomgen.s3.amazonaws.com/latest/linux/amd64/inspector-sbomgen.zip (https://amazon-inspector-sbomgen.s3.amazonaws.com/latest/linux/amd64/inspector-sbomgen.zip)

   Linux ARM64: https://amazon-inspector-sbomgen.s3.amazonaws.com/latest/linux/arm64/inspector-sbomgen.zip (https://amazon-inspector-sbomgen.s3.amazonaws.com/latest/linux/arm64/inspector-sbomgen.zip)

   Alternatively, you can download previous versions of the Amazon Inspector SBOM Generato file (https://docs.aws.amazon.com/inspector/latest/user/sbom-generator-versions.html) .
2. Unzip the download using the following command:

   ```
   unzip inspector-sbomgen.zip
   ```
3. Check for the following files in the extracted directory:
   - `inspector-sbomgen` – This is the tool you will execute to generate SBOMs.
   - `README.txt` – This is the documentation for using Sbomgen.
   - `LICENSE.txt` – This file contains the software license for Sbomgen.
   - `licenses` – This folder contains license info for third party packages used by Sbomgen
   - `checksums.txt` – This file provides hashes of the Sbomgen tool.
   - `sbom.json` – This is a CycloneDX SBOM for the Sbomgen tool.
   - `WhatsNew.txt` – This file contains a summarized change log, so you can view major changes and improvements between Sbomgen versions quickly.
4. (Optional) Verify the authenticity and integrity of the tool using the following command:

   ```
   sha256sum < inspector-sbomgen
   ```
   a. Compare the results with the contents of the `checksums.txt` file.
5. Grant executable permissions to the tool using the following command:

   ```
   chmod +x inspector-sbomgen
   ```
6. Verify that Sbomgen is successfully installed using the following command:

   ```
   ./inspector-sbomgen --version
   ```

   You should see the output similar to the following:

   ```
   Version: 1.X.X
   ```

Using Sbomgen

This section describes different ways you can use Sbomgen. You can learn more about how to us
Sbomgen through built-in examples. To view these examples, run the `list-examples` comma

```
./inspector-sbomgen list-examples
```

Generate an SBOM for a container image and output the result

You can use Sbomgen to generate SBOMs for container images and output the result to a file. Tl
capability can be enabled using the `container` subcommand.

Example command

In the following snippet, you can replace *`image:tag`* with the ID of your image and *`output_path.json`* with the path to the output you want to save.

```
# generate SBOM for container image
./inspector-sbomgen container --image image:tag -o output_path.json
```

> **Note**
>
> Scan time and performance depends on the image size and how small the number of laye
> are. Smaller images not only improve Sbomgen performance, but also reduce the potenti
> attack surface. Smaller images also improve image build, download, and upload times.

When using Sbomgen with ScanSbom (https://docs.aws.amazon.com/inspector/v2/APIReference/API_scan_ScanSbom.html) , the Amazon Insp
Scan API won't process SBOMs that contain more than 5,000 packages. In this scenario, the Ama
Inspector Scan API returns an HTTP 400 response.

If an image includes bulk media files or directories, consider excluding them from Sbomgen usin
`--skip-files` argument.

Example: Common error cases

Container image scanning can fail due to the following errors:

- `InvalidImageFormat` – Occurs when scanning malformed container images with corrupt
  headers, manifest files, or config files .
- `ImageValidationFailure` – Occurs when checksum or content-length validation fails fo
  container image components, such as mismatched Content-Length headers, incorrect manif
  digests, or failed SHA256 checksum verification.
- `ErrUnsupportedMediaType` – Occurs when image components include unsupported mec
  types. For information about supported media types, see Supported operating systems and
  media types (https://docs.aws.amazon.com/inspector/latest/user/scanning-ecr.html#ecr-supported-
  media) .

Amazon Inspector does not support the
`application/vnd.docker.distribution.manifest.list.v2+json` media type. Howev
Amazon Inspector does support manifest lists. When scanning images that use manifest lists, yo
explicitly specify which platform to use with the `--platform` argument. If the `--platform`

argument is not specified, the Amazon Inspector SBOM Generator automatically selects the mar based on the platform where its running.

Generate an SBOM from directories and archives

You can use Sbomgen to generate SBOMs from directories and archives. This capability can be enabled using the `directory` or `archive` subcommands. Amazon Inspector recommends usi this feature when you want to generate an SBOM from a project folder, such as a downloaded gi repository.

Example command 1

The following snippet shows a subcommand that generates an SBOM from a directory file.

```
# generate SBOM from directory
./inspector-sbomgen directory --path /path/to/dir -o /tmp/sbom.json
```

Example command 2

The following snippet shows a subcommand that generates an SBOM from an archive file. The o supported archive formats are `.zip`, `.tar`, and `.tar.gz`.

```
# generate SBOM from archive file (tar, tar.gz, and zip formats only)
./inspector-sbomgen archive --path testData.zip -o /tmp/sbom.json
```

Generate an SBOM from Go or Rust compiled binaries

You can use Sbomgen to generate SBOMs from compiled Go and Rust binaries. You can enable t cabapility through the `binary` subcommand:

```
./inspector-sbomgen binary --path /path/to/your/binary
```

Generate an SBOM from mounted volumes

You can use Amazon Inspector SBOM Generator to generate SBOMs from mounted volumes. Thi capability can be enabled using the `volume` subcommand. We recommend using this feature w you want to analyze storage volumes, such as Amazon EBS volumes that have been mounted to system. Unlike the directory subcommand, mounted volume scanning detects OS packages and information.

You can scan an Amazon EBS volume by attaching it to an Amazon EC2 instance where Amazon Inspector SBOM Generator is installed and mounting it on that instance. For Amazon EBS volum that are currently in use by other Amazon EC2 instances, you can create an Amazon EBS snapsho the volume and then create a new Amazon EBS volume from that snapshot for scanning purpos For more information about Amazon EBS, see What is Amazon EBS? (https://docs.aws.amazon.com/ebs/latest/userguide/what-is-ebs.html) in the *Amazon Elastic Block Stor User Guide*.

Example command

The following snippet shows a subcommand that generates an SBOM from a mounted volume. T `-path` argument should specify the root directory where the volume is mounted.

```
# generate SBOM from mounted volume
./inspector-sbomgen volume --path /mount/point/of/volume/root
```

Example command

The following snippet shows a subcommand that generates an SBOM from a mounted volume v
excluding specific file paths with the `--exclude-suffix` argument. The `--exclude-suffix`
argument is particularly useful when a volume contains bulk files (such as log files or media files
Files and directories whose paths end with the specified suffixes will be excluded from scanning,
which can reduce scan time and memory usage.

```
# generate SBOM from mounted volume with exclusions
./inspector-sbomgen volume --path /mount/point/of/volume/root \
--exclude-suffix .log \
--exclude-suffix cache
```

All file paths in the target volume are normalized to their original paths. For example, when scar
a volume mounted at `/mnt/volume` that contains a file at
`/mnt/volume/var/lib/rpm/rpmdb.sqlite`, the path will be normalized to
`/var/lib/rpm/rpmdb.sqlite` in the generated SBOM.

Send an SBOM to Amazon Inspector for vulnerability identification

In addition to generating an SBOM, you can send an SBOM for scanning with a single command
the Amazon Inspector Scan API. Amazon Inspector evaluates the contents of the SBOM for
vulnerabilites before returning findings to Sbomgen. Depending on your input, the findings can
displayed or written to a file.

> **Note**
>
> You must have an active AWS account with read permissions to `InspectorScan-ScanSbom` to use this capability.

To enable this capability, you pass the `--scan-sbom` argument to the Sbomgen CLI. You can al
pass the `--scan-sbom` argument to any of the following Sbomgen subcommands: `archive`,
`binary`, `container`, `directory`, `localhost`.

> **Note**
>
> The Amazon Inspector Scan API doesn't process SBOMs with more than 5,000 packages. I
> this scenario, the Amazon Inspector Scan API returns an HTTP 400 response.

You can authenticate to Amazon Inspector through an AWS profile or an IAM role with the follov
AWS CLI arguments:

```
--aws-profile profile
--aws-region region
--aws-iam-role-arn role_arn
```

You can also authenticate to Amazon Inspector by providing the following environment variable
Sbomgen.

```
AWS_ACCESS_KEY_ID=$access_key \
AWS_SECRET_ACCESS_KEY=$secret_key \
AWS_DEFAULT_REGION=$region \
./inspector-sbomgen arguments
```

To specify the response format, use the `--scan-sbom-output-format cyclonedx` argumer `--scan-sbom-output-format inspector` argument.

Example command 1

This command creates an SBOM for the latest Alpine Linux release, scans the SBOM, and writes t vulnerability results to a JSON file.

```
./inspector-sbomgen container --image alpine:latest \
                          --scan-sbom \
                          --aws-profile your_profile \
                          --aws-region your_region \
                          --scan-sbom-output-format cyclonedx \
                          --outfile /tmp/inspector_scan.json
```

Example command 2

This command authenticates to Amazon Inspector using AWS credentials as environment variabl

```
AWS_ACCESS_KEY_ID=$your_access_key \
AWS_SECRET_ACCESS_KEY=$your_secret_key \
AWS_DEFAULT_REGION=$your_region \
./inspector-sbomgen container --image alpine:latest \
                          -o /tmp/sbom.json \
                          --scan-sbom \
                          --scan-sbom-output-format inspector
```

Example command 3

This command authenticates to Amazon Inspector using the ARN for an IAM role.

```
./inspector-sbomgen container --image alpine:latest \
                          --scan-sbom \
                          --aws-profile your_profile \
                          --aws-region your_region \
                          --outfile /tmp/inspector_scan.json
                          --aws-iam-role-arn
arn:aws:iam::123456789012:role/your_role
```

Use additional scanners to enhance detection capabilities

The Amazon Inspector SBOM Generator applies predefined scanners based on the command bei used.

Default scanner groups

Each Amazon Inspector SBOM Generator subcommand applies the following default scanner gro automatically.

- For the `directory` subcommand: binary, programming-language-packages, dockerfile sca groups
- For the `localhost` subcommand: os, programming-language-packages, extra-ecosystems scanner groups
- For the `container` subcommand: os, programming-language-packages, extra-ecosystems, dockerfile, binary scanner groups

Special scanners

To include scanners beyond the default scanner groups, use the `--additional-scanners` op followed by the name of the scanner to be added. The following is an example command showir how to do this.

```
# Add WordPress installation scanner to directory scan
./inspector-sbomgen directory --path /path/to/directory/ --
additional-scanners wordpress-installation -o output.json
```

The following is an example command showing how to add multiple scanners with a comma-separated list.

```
./inspector-sbomgen container --image image:tag --additional-scanners
scanner1,scanner2 -o output.json
```

Optimize container scans by adjusting maximum file size to scan

When you analyze and process a container image, Sbomgen scans files that are 200 MB or less b default. Files larger than 200 MB rarely contain package metadata. You can encounter misses wh you inventory a Go or Rust binary that exceeds 200MB. To adjust the size limit, use the `--max-file-size` argument. This allows you to increase the limit to include large files and decrease tl limit to reduce resource usage by excluding large files.

Example

The following example shows how to use the `--max-file-size` argument to increase the file

```
# Increase the file size limit to scan files up to 300 MB
./inspector-sbomgen container --image alpine:latest \
--outfile /tmp/sbom.json \
--max-file-size 300000000
```

Adjusting this setting helps control disk usage, memory consumption, and overall scan duration.

Disable progress indicator

Sbomgen displays a spinning progress indicator that can result in excessive slash characters in CI environments.

```
INFO[2024-02-01 14:58:46]coreV1.go:53: analyzing artifact
|
\
/
|
\
/
INFO[2024-02-01 14:58:46]coreV1.go:62: executing post-processors
```

You can disable the progress indicator using the `--disable-progress-bar` arguement:

```
./inspector-sbomgen container --image alpine:latest \
--outfile /tmp/sbom.json \
--disable-progress-bar
```

Authenticating to private registries with Sbomgen

By providing your private registry authentication credentials, you can generate SBOMs from containers that are hosted in private registries. You can provide these credentials through the following methods:

Authenticate using cached credentials (recommended)

For this method, you authenticate to your container registry. For example, if using Docker, you ca authenticate to your container registry using the Docker loging command: `docker login`.

1. Authenticate to your container registry. For example, if using Docker, you can authenticate t your registry using the Docker `login` command:
2. After you authenticate to your container registry, use Sbomgen on a container image that's i registry. To use the following example, replace *`image:tag`* with the name of the image to

```
./inspector-sbomgen container --image image:tag
```

Authenticate using the interactive method

For this method, provide your username as a parameter, and Sbomgen will prompt you for secur password entry when needed.

To use the following example, replace *`image:tag`* with the name of the image that you want t scan and *`your_username`* with a username that has access to the image:

```
./inspector-sbomgen container --image image:tag --username
your_username
```

Authenticate using the non-interactive method

For this method, store your password or registry token in a `.txt` file.

Note

The current user should only be able to read this file. The file should also contain your password or token on a single line.

To use the following example, replace *your_username* with your username, *password.txt* the `.txt` file that includes your password or token on a single line, and *image:tag* with the of the image to scan:

```
INSPECTOR_SBOMGEN_USERNAME=your_username \
INSPECTOR_SBOMGEN_PASSWORD=`cat password.txt` \
./inspector-sbomgen container --image image:tag
```

Example outputs from Sbomgen

The following is an example of an SBOM for a container image inventoried using Sbomgen.

Container image SBOM


```
{
  "bomFormat": "CycloneDX",
  "specVersion": "1.5",
  "serialNumber": "urn:uuid:828875ef-8c32-4777-b688-
0af96f3cf619",
  "version": 1,
  "metadata": {
    "timestamp": "2023-11-17T21:36:38Z",
    "tools": [
      {
        "vendor": "Amazon Web Services, Inc. (AWS)",
        "name": "Amazon Inspector SBOM Generator",
        "version": "1.0.0",
        "hashes": [
          {
            "alg": "SHA-256",
            "content":
"10ab669cfc99774786301a745165b5957c92ed9562d19972fbf344d4393b5eb1
"
          }
        ]
      }
    ],
    "component": {
      "bom-ref": "comp-1",
      "type": "container",
      "name": "fedora:latest",
      "properties": [
        {
          "name": "amazon:inspector:sbom_generator:image_id",
          "value":
```

```
"sha256:c81c8ae4dda7dedc0711daefe4076d33a88a69a28c398688090c1141e
ff17e50"
        },
        {
          "name":
"amazon:inspector:sbom_generator:layer_diff_id",
          "value":
"sha256:eddd0d48c295dc168d0710f70364581bd84b1dda6bb386c4a4de0b61d
e2f2119"
        }
      ]
    }
  },
  "components": [
    {
      "bom-ref": "comp-2",
      "type": "library",
      "name": "dnf",
      "version": "4.18.0",
      "purl": "pkg:pypi/dnf@4.18.0",
      "properties": [
        {
          "name":
"amazon:inspector:sbom_generator:source_file_scanner",
          "value": "python-pkg"
        },
        {
          "name":
"amazon:inspector:sbom_generator:source_package_collector",
          "value": "python-pkg"
        },
        {
          "name": "amazon:inspector:sbom_generator:source_path",
          "value": "/usr/lib/python3.12/site-packages/dnf-
4.18.0.dist-info/METADATA"
        },
        {
          "name":
"amazon:inspector:sbom_generator:is_duplicate_package",
          "value": "true"
        },
        {
          "name":
"amazon:inspector:sbom_generator:duplicate_purl",
          "value": "pkg:rpm/fedora/python3-dnf@4.18.0-2.fc39?
arch=noarch&distro=39&epoch=0"
        }
      ]
    },
    {
      "bom-ref": "comp-3",
      "type": "library",
      "name": "libcomps",
      "version": "0.1.20",
      "purl": "pkg:pypi/libcomps@0.1.20",
```

```
      "properties": [
        {
          "name":
"amazon:inspector:sbom_generator:source_file_scanner",
          "value": "python-pkg"
        },
        {
          "name":
"amazon:inspector:sbom_generator:source_package_collector",
          "value": "python-pkg"
        },
        {
          "name": "amazon:inspector:sbom_generator:source_path",
          "value": "/usr/lib64/python3.12/site-packages/libcomps-
0.1.20-py3.12.egg-info/PKG-INFO"
        },
        {
          "name":
"amazon:inspector:sbom_generator:is_duplicate_package",
          "value": "true"
        },
        {
          "name":
"amazon:inspector:sbom_generator:duplicate_purl",
          "value": "pkg:rpm/fedora/python3-libcomps@0.1.20-
1.fc39?arch=x86_64&distro=39&epoch=0"
        }
      ]
    }
  ]
}
```


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